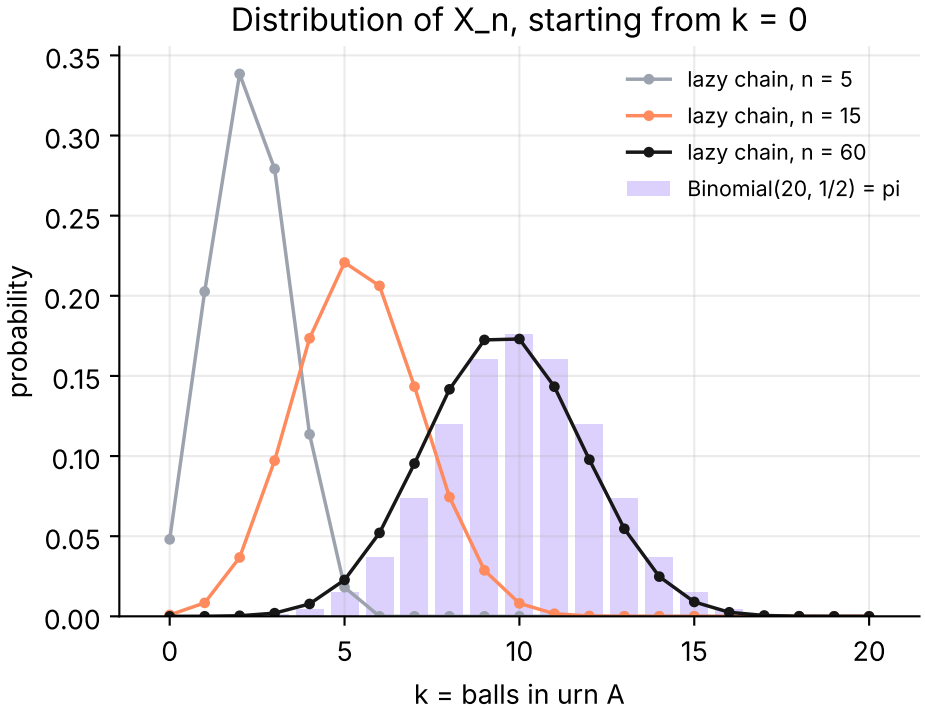
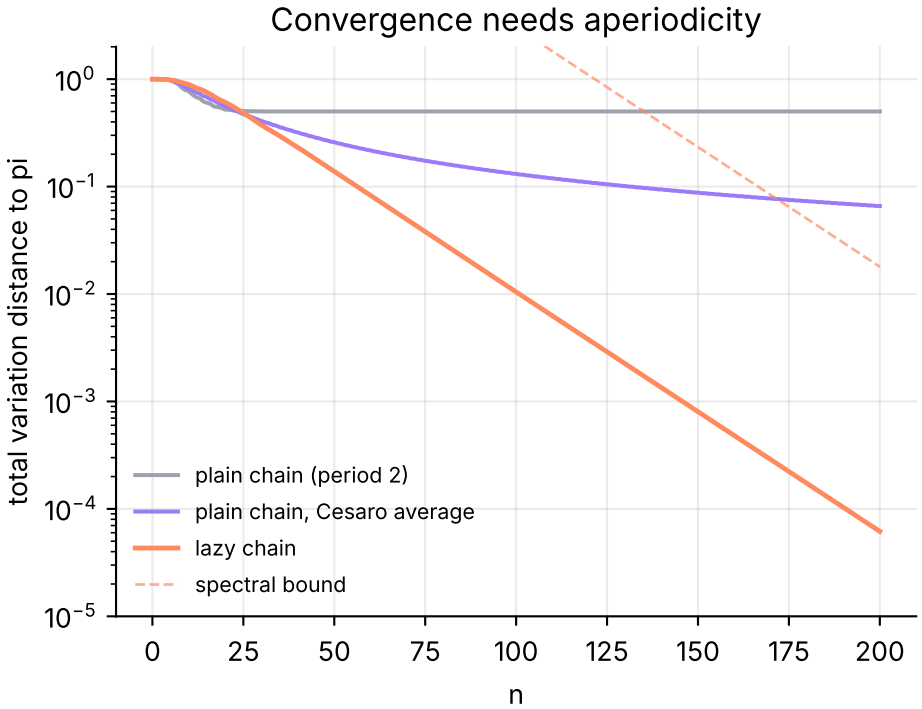

Distribution of X_n, starting from k = 0
probability
k = balls in urn A
lazy chain, n = 5
lazy chain, n = 15
lazy chain, n = 60
Binomial(20, 1/2) = pi
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0
5
10
15
20


Convergence needs aperiodicity
total variation distance to pi
n
plain chain (period 2)
plain chain, Cesaro average
lazy chain
spectral bound
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
25
50
75
100
125
150
175
200